Grade: Check Plus

Assignment 1 ✓ Qa

a) 0.15 + 0.25 + 0.05 = 0.45 or 45%

b) (10*0.1) + (20*0.15) + (40*0.30) + (60*0.15) + (80*0.25) + (100*0.05) = $50

	x	f(x)	x*f(x)	Variance (x-50)^2*f(x)	
	10	0.1	1	160	
	20	0.15	3	135	
	40	0.3	12	30	
	60	0.15	9	15	
	80	0.25	20	225	
	100	0.05	5	125	
Sum	310	1	**50**	**690**	Variance
				26.27	Standard Deviation

✓ Qb

✓ Qc

c) $50 is the theoretical average (mean) that we expect to happen in the long run and therefore it doesn't matter that it is not reflected as a possible withdrawal amount

d) P(50-26.27<=X<=50+26.27) = ✓ Qd
P(23.73<=X<=76.27) =
P(24<=X<=76) = P(40) + P(60) = 0.45 or 45%

e) 50-2*26.27<=X<=50+2*26.27) = ✓ Qe
P(-2.54<=X<=102.54) =
P(-2<=X<=102) = P(10) +P(20) + P(40) + P(60) + P(80) + P(100) = 1.0 or 100%